

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 6+ years shipping backend systems in production — APIs, automation pipelines, and cloud infrastructure — on Node.js/TypeScript and Python across GCP (Cloud Functions, Firestore, Firebase, Pub/Sub) and AWS. Led the end-to-end, zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP at MTailor, reporting directly to the CTO, and built the real-time Pub/Sub sync pipeline that kept two datastores consistent under live traffic. Owns features from problem to production and iterates on real usage: Express Shipping added \$40 revenue per order; a 3D visualisation feature lifted buyer conversion. Built services for users on feature phones (USSD legal aid, Uganda) and DeFi backends on Web3 — now bringing the same end-to-end ownership to financial products for underbanked workers.

CORE COMPETENCIES

Node.js / TypeScript

Python

GCP: Cloud Functions, Firestore, Firebase, Pub/Sub

AWS (S3, EC2, EBS)

REST API Design

Serverless & Event-Driven Pipelines

Docker & Kubernetes

End-to-End Feature Ownership

WORK EXPERIENCE

MTailor · US Remote Sep 2022 – Present

Software Engineer — Backend (reporting to the CTO)

- Led the end-to-end migration of 20+ production applications from Parse/MongoDB to Firebase/GCP with **zero downtime**, owning design, cutover sequencing, and rollback for every service
- Built a real-time two-way sync pipeline between MongoDB and Firestore in Node.js on Google Pub/Sub, keeping both datastores consistent while they served live production traffic
- Shipped Express Shipping end-to-end — backend, pricing logic, and ops handoff — generating an additional **\$40 revenue per order**
- Built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion
- Consolidated cloud infrastructure onto GCP: migrated file storage from Amazon S3 to Google Cloud Storage via a Python script and moved all remaining services off AWS, saving **\$5,000/month**
- Migrated the Webflow marketing site from a Parse backend to Firebase

MTailor · US Remote · Contract Mar 2022 – Jul 2022

Backend Engineer (Contractor)

- Built the MongoDB → Firestore migration scripts, applying complex data-processing logic during transfer; converted to full-time and given ownership of the company-wide migration
- Wrote documentation for the Firebase SDKs (Node.js and Python) that fellow engineers onboarded from; trained the Ops team on the Firebase dashboard and backup procedures

Mind2matter.co · US Remote Jan 2022 – Jul 2022

Software Engineer — Full-Stack

- Built backends for DeFi applications using Web3 and Node.js for a US agency with demanding clients, reporting directly to the CTO
- Delivered React UIs from Figma designs under tight timelines

Dr Wealth · Singapore Remote Aug 2021 – Jan 2022

Software Engineer — Backend

- Extended backend APIs using Firebase Cloud Functions and Express.js (hosted on Heroku) for a PWA delivering real-time stock market prices
- Built automated jobs querying 2M+ Firestore records to keep customer prices current from Morningstar APIs, optimising query shape over hardware
- Added responsive UI pages to the PWA with React and Tailwind

Software Engineer & Team Lead (4 developers)

- Architected an event-driven microservices backend for FIDA Uganda's case-management system using Apache Kafka, Docker, and Kubernetes; led the team through delivery
- Built a USSD legal-aid service for feature-phone users communicating with a Node.js backend via gRPC, with a provider-agnostic contract multiple legal-aid providers integrated against
- Built a paralegal database for LASPNET Uganda and a cross-platform React Native/Expo mobile app (iOS & Android) with Firebase push notifications

SELECTED PROJECTS

Real-time MongoDB ↔ Firestore Sync Pipeline Production

Idempotent, redelivery-safe two-way sync on Google Pub/Sub that let two datastores serve parallel production traffic for the full duration of a 20+ app migration — the backbone of the zero-downtime cutover.

Node.js · Google Pub/Sub · Firestore · MongoDB

USSD Legal-Aid Service for Rural Uganda Underserved users

Legal-aid access over basic phones for citizens without smartphones or data; gRPC contract with explicit versioning so multiple providers integrated without bespoke code.

Node.js · gRPC · Kafka · USSD

Healthcare WhatsApp Chatbot with Prescription OCR AI in production

Conversational assistant on the WhatsApp Business API integrating OpenAI for dialogue and Google Vision for prescription OCR, handling media ingestion and platform rate limits.

Node.js · OpenAI · Google Vision · WhatsApp Business API

Marketplace Recommendation Engine Data modelling

Collaborative + content-based filtering engine in MongoDB ranking service providers from messy, user-generated catalog data.

Node.js · MongoDB

EDUCATION

BSc Computer Engineering — Makerere University, Kampala

2018 – 2022

SKILLS

Languages: Node.js, TypeScript, JavaScript, Python · Go, Rust (intermediate)

Cloud: GCP (Cloud Functions, Firestore, Firebase, Pub/Sub, GCS), AWS (S3, EC2, EBS)

Backend: REST APIs, Express.js, serverless, event-driven pipelines, gRPC, Apache Kafka, NATS Streaming

Infrastructure: Docker, Kubernetes, Linux, Git **Data:** Firestore, MongoDB, SQL **Frontend:** React, React Native, Tailwind, HTML/CSS

Spoken: English (fluent), Kiswahili